

Roof Runoff and Gutter Bypass Analysis

1. Problem Summary

This report addresses the issue of water from an upper roof bypassing the gutter system of a lower, adjacent roof. This problem, often referred to as "overshooting," leads to improper water drainage, potential water damage to the building's facade, and erosion of the ground below. The analysis of the provided image and video confirms that the velocity and volume of water coming from the steep upper roof are causing it to flow over the lower gutter, especially during moderate to heavy rainfall.

2. Analysis of Provided Media

Still Image Description (90-1.png)

The image displays a two-level roof structure. The upper roof has a noticeably steep pitch, directing a significant amount of water towards the lower roof. The lower roof has a shallower angle and is equipped with a standard gutter system. The positioning of the gutter appears to be correct, but it is not equipped to handle the concentrated flow from the upper roof valley. The water stains on the fascia and siding below the gutter line are visible indicators of persistent overflow.

Video Summary (90-2.mp4)

The video footage captures the dynamic behavior of the roof runoff during a rain event. Key observations include:

- Water flows rapidly down the upper roof.
- Upon reaching the edge of the upper roof, the water is projected outwards, clearing the lower gutter entirely.
- A significant splash point is observable on the ground and lower wall section, confirming the bypass.
- The lower gutter only collects water that falls directly onto the lower roof section, not the runoff from the upper level.

3. Roof Slope Calculation

A precise calculation of the upper roof's slope angle is not feasible without direct measurements or architectural plans. However, based on visual analysis, the pitch

appears to be steep, likely in the range of 8/12 to 12/12 (approximately 33 to 45 degrees). This steepness is a major contributing factor to the water's velocity, causing it to overshoot the gutter.

4. Proposed Solution: Gutter Modifications

Research into standard roofing and gutter installation practices for multi-level roofs indicates that this is a common issue. The most effective solutions involve installing physical barriers to redirect the water flow.

Recommended Solutions:

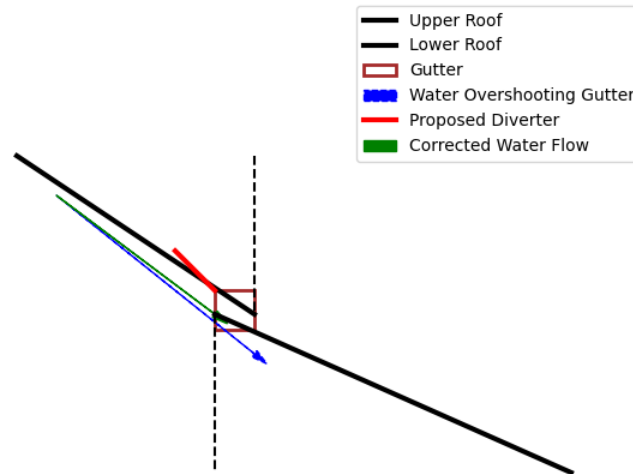
1. ****Install a Diverter or Splash Guard:**** This is the most direct solution. A diverter, typically an L-shaped piece of metal flashing, is installed at the edge of the upper roofline to interrupt the water's trajectory and direct it downwards into the gutter.
2. ****Install a Larger Gutter:**** While a standard 5-inch gutter may be overwhelmed, upgrading to a 6-inch gutter with a wider opening can help capture more of the runoff.
3. ****Adjust Gutter Position:**** Ensuring the gutter is positioned as high and close to the roofline as possible can minimize the chance of overshoot.

The most cost-effective and targeted solution is the installation of a diverter.

5. Annotated Schematic of the Roof-Gutter System

The following diagram illustrates the current problem and the proposed solution.

Roof Cross-Section and Proposed Gutter Modification



Legend:

- **Blue Dashed Arrow:** Shows the current path of water, overshooting the gutter.
- **Red Line:** Represents the proposed diverter/splash guard.
- **Green Arrow:** Shows the corrected water flow into the gutter after the modification.

6. Step-by-Step Implementation Instructions

Objective: Install a roof diverter to guide water from the upper roof into the lower gutter.

Materials:

- Sheet metal (aluminum or galvanized steel, L-shaped) or a pre-fabricated roof diverter.
- Roofing-grade sealant.
- Appropriate screws with rubber washers.
- Drill, measuring tape, and safety equipment.

Instructions:

1. **Safety First:** Use a sturdy, properly positioned ladder. Ensure you have all

necessary safety gear, including gloves and eye protection.

2. ****Measure and Cut the Diverter:**** Measure the length of the roof edge where the overflow is occurring. Cut the diverter to this length. The vertical flange of the "L" should be at least 2-3 inches high.

3. ****Clean the Area:**** Thoroughly clean the roof surface and the edge of the shingles where the diverter will be placed. Remove any leaves, dirt, or debris.

4. ****Apply Sealant:**** Apply a continuous bead of roofing sealant along the underside of the diverter's top flange (the part that will sit on the roof).

5. ****Position the Diverter:**** Slide the top flange of the diverter under the edge of the roof shingles. The vertical flange should be positioned to block the water's path and redirect it into the gutter.

6. ****Secure the Diverter:**** Fasten the diverter to the roof deck using the rubber-washer screws. Place screws every 12-18 inches. Do not overtighten, as this can damage the shingles.

7. ****Seal the Edges:**** Apply another bead of sealant along the top edge of the diverter where it meets the shingles to create a watertight seal.

8. ****Test the System:**** If possible, use a hose to simulate rainfall and check that the water is now being correctly diverted into the gutter.